Special Topics for EEE

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Assignment 2:

(1)

$$\theta = 1.22 \frac{500 \times 10^{-9}\,nm}{8 \times 10^{-2}\,nm} = 7.625 \times 10^{-6}\,rad$$

(2)

$$x_n = \frac{n \lambda L}{a}$$

$a = 100\,\mu m$
$L = 5\,m$

$x_0 = 0 \times 460\,nm \times 5m / 100\,\mu m = 0$
$x_1 = 1 \times 460\,nm \times 5m / 100\,\mu m = 0.23\,nm$
$x_2 = 2 \times 460\,nm \times 5m / 100\,\mu m = 0.46\,nm$

(4)

$$r_{TE} = \frac{n_1 \cos\theta_i - n_2 \cos\theta_t}{n_1 \cos\theta_i + n_2 \cos\theta_t}, \quad r_{TE} = \frac{n_1 \cos\theta_i - n_2 \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2}}{n_1 \cos\theta_i + n_2 \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2}}$$

$$T = \frac{n_2 \cos\theta_t}{n_1 \cos\theta_i} |t_{TE}|^2 = 0.95$$

$$t_{TE} = \frac{2 n_1 \cos\theta_i}{n_1 \cos\theta_i + n_2 \cos\theta_t}, \quad t_{TE} = \frac{2 n_1 \cos\theta_i}{n_1 \cos\theta_i + n_2 \sqrt{1 - \left(\frac{n_1}{n_2}\sin\theta_i\right)^2}}$$

$$\theta_i = \arcsin\left(\frac{n_2}{n_1}\sqrt{\frac{0.95}{2}}\right) = 41.4$$

$$\sin\theta_t = \frac{n_1}{n_2}\sin\theta_i$$

(5)

$I = I_0 \cos^2\theta$
$\Delta\theta = 30^\circ - 0^\circ = 30^\circ$

$I_1 = 800 \times \cos 30 = 600\ w/m^2$
$I_2 = I_1 \cos^2\theta \Rightarrow 600 \cdot \cos^2 30 = 450\ w/m^2$

$\theta = 45^\circ$
$I_0 = 225\ w/m^2$

(6) —

(7)

$$F_1 = \frac{2I}{c} = \frac{2 \cdot 1350}{3 \times 10^8} = 9 \times 10^{-6}\,A \qquad \frac{F}{mg} = \frac{9 \times 10^{-6}}{9.8} = 9.18 \times 10^{-7}\,A$$